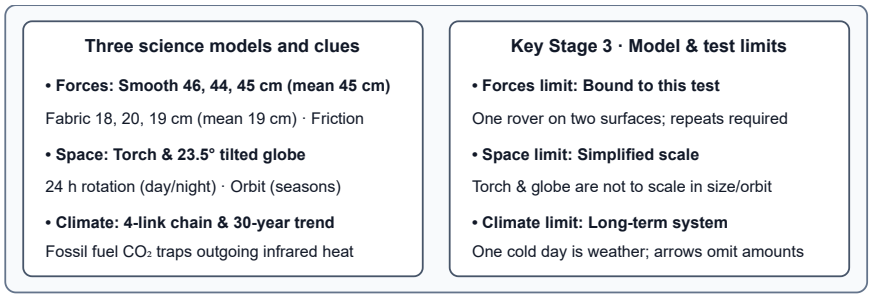
Reference sheets

Reference A · Three science models, clues, reasons, and limits

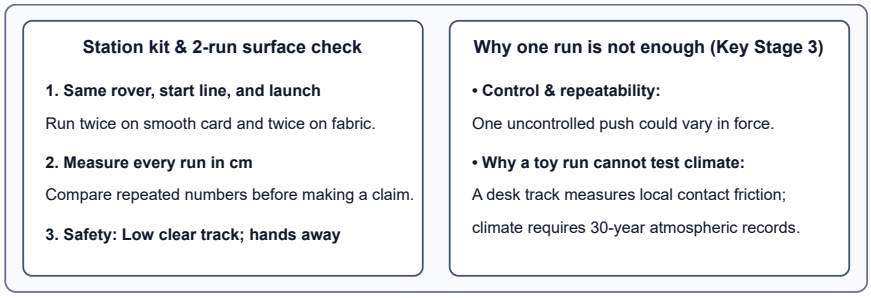


Three science models diagram. It shows Forces surface runs, Space torch and globe, and Climate warming chain with limits.

Reference only. Do not write on this sheet. Read the question first, then pick the matching model row.

Science question.	Model and evidence clue.	Key Stage 3 science reason.	Model or test limit.
1. Forces: Which surface lets the rover travel farther?	Rover track: Smooth 46, 44, 45 cm (mean 45 cm). Fabric 18, 20, 19 cm (mean 19 cm).	Contact friction opposes motion between rover wheels and the surface.	Only one rover and two surfaces were tested ('in this test').
2. Space: Why is a location in darkness or winter?	Fixed torch and turning 23.5°-tilted globe: marked spot turns into shadow.	Earth's 24-hour rotation causes day and night. Its 23.5° tilt during orbit causes seasons.	The classroom torch and globe are not to scale ('in this model').
3. Climate: Why does extra CO_2 warm Earth over decades?	Four-link warming chain and 30-year climate trend.	Fossil fuel CO_2 traps outgoing infrared heat. Mitigation cuts gas (Link 1); adaptation cuts harm (Link 4).	One cold day is weather, not climate. Arrows omit exact amounts.

Reference B · Desk station check and safe two-run rover method



Desk station check diagram. It shows the two-run rover method, safety rules, and test limits.

Station step.	What we do at the desk.	Key Stage 3 reason and safety rule.
1. Match kit to question.	Use rover + surfaces for motion, torch + globe for day/night, and warming cards for climate.	A model can only test a question that matches its mechanism.
2. Fair two-run rover check.	Keep the same rover, start line, and push. Measure two runs per surface in cm.	One uncontrolled run cannot prove a cause. Repeats check if the pattern holds.
3. Safety on the desk.	Use a low clear track. Keep hands and faces away. Keep torch light out of eyes.	Prevents the moving toy falling or hitting anyone.

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use the arrival diagram. Look at the true fact 'Earth rotates in 24 hours'. Why is it wrong as rover evidence?

Help. Check if the mechanism matches the question.

2. In Box 1, smooth trials are 46, 44, and 45 cm. Fabric trials are 18, 20, and 19 cm. Compare the means.

Help. Add the three trials and divide by 3 for each surface.

3. State one Key Stage 3 limit of the globe model in Box 2. State one limit of the warming arrows in Box 3.

Help. Compare the classroom model with the real Solar System and atmosphere.

4. Why must we bound a surface claim with 'in this test'? Why not say 'on every surface everywhere'?

Help. Keep the claim inside what was actually measured.

You do tasks

1. Use Reference A and Reference B. Compare all three models (surface distance table, torch and 23.5°-tilted globe, and warming chain). State the Key Stage 3 mechanism and one limit for each. **Staff I ☐ S ☐ C ☐**

Help. Link contact friction (means 45 cm vs 19 cm), 24-hour rotation + 23.5° axial tilt, and CO_2 infrared trapping + mitigation vs adaptation.

2. A pupil says: 'My toy rolled 46 cm on smooth paper once. Every vehicle everywhere goes faster on smooth ground.' Repair this claim. **Staff I ☐ S ☐ C ☐**

Help. Check for repeats, controls, and bounding the claim to this test.

3. What extra evidence would strengthen the rover surface claim? Why can a toy track never test a 30-year climate claim? **Staff I ☐ S ☐ C ☐**

Help. Compare repeated controlled trials on the desk with 30-year atmospheric records.

Optional participation-loop prompt. Share your model match and clue with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Look at the true statement 'Earth rotates every 24 hours'. Why is it irrelevant evidence for the rover surface table?

2. Use the practice means (45 cm vs 19 cm) and contact friction. Write a bounded claim for the rover test.

3. Compare one model limit in Space (torch and 23.5°-tilted globe) with one model limit in Climate (warming-chain arrows).

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt